

Egypt Water Quality Analysis & Prediction System

AI-Powered Water Quality Monitoring
and Prediction Platform

Project Team



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System Workflow

The Data Journey: From raw measurement to actionable intelligence



System Architecture



Power BI Analytics Dashboard

Real-Time Visualization

Mapping live water quality metrics against geographical and temporal data.



Trend Analysis

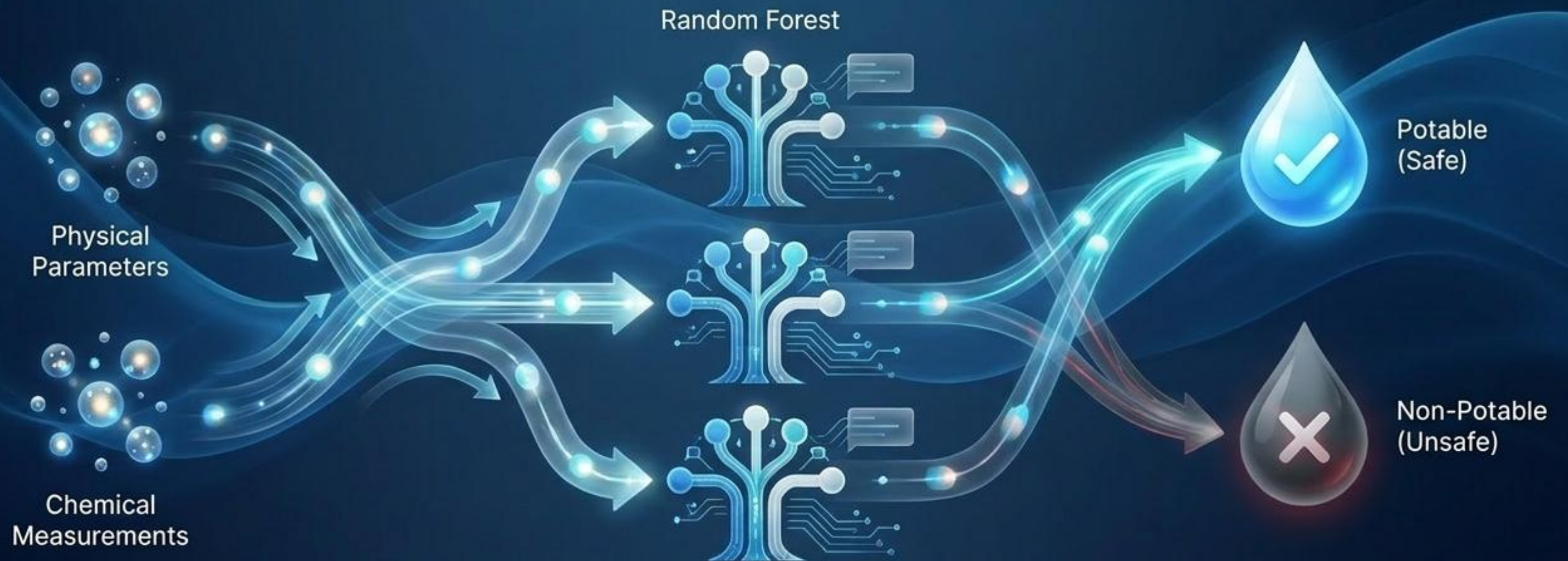
Tracking historical fluctuations in physical and chemical parameters.

KPI Monitoring

Instant visibility into critical thresholds and automated deviations.

AI Detection Model

Random Forest Classifier



Future Enhancements

National Platform

A unified, country-wide water monitoring network.



Cloud Deployment

Seamless scalability and remote hosting.



Mobile App

On-the-go accessibility for field workers.



Automated Alerts

Push notifications for critical contamination events.



IoT Integration

Direct streaming from hardware sensors.





Securing Egypt's Water Future

Transforming raw environmental data into life-saving intelligence.